SNMP Lab Documentation

Cisco Router Monitoring and Security Hardening

End-to-end implementation of SNMPv1/v2c operations and SNMPv3 secure management

Scope	SNMP deployment, monitoring, trap handling, and protocol analysis
Platform	GNS3 + Cisco IOS + SnmpB + Wireshark + Windows host tools
Network Segment	172.16.0.0/24 (Manager: 172.16.0.2, Agent: 172.16.0.1)
Security Focus	SNMPv3 authPriv model with SHA authentication and DES privacy
Deliverable	Reproducible lab procedure with validation evidence

April 21, 2026

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1. Project Overview

This lab documents a complete SNMP lifecycle across two phases:

- **Operational phase (SNMPv1/v2c):** manager-agent connectivity, MIB walks, and trap-driven monitoring.
- **Security phase (SNMPv3):** authenticated and encrypted SNMP with role-based access control.

The objective is to build a practical monitoring setup, observe real packet behavior, and validate how security controls affect both protocol operation and traffic visibility.

- Practical configuration steps with command-level detail.
- Validation through SnmpB outputs and Wireshark captures.
- Comparison of SNMP versions from both functional and security perspectives.

2. Lab Environment and Topology

2.1. Toolchain

Tool	Purpose
WinPcap	Packet capture driver used by Wireshark on Windows
Wireshark	Deep packet analysis and SNMP protocol dissection
GNS3	Emulation of Cisco router topology
SnmpB	SNMP manager for polling, walking MIBs, and viewing traps

2.2. Addressing and Roles

- **SNMP Manager Host (Loop1):** 172.16.0.2/24
- **Cisco Router R1 (f0/0):** 172.16.0.1/24
- **Test Interface for Traps:** R1 f1/0 (administrative up/down events)

2.3. Host-Side Network Preparation

PowerShell – Configure Loopback Adapter

```
$adapter = Get-NetAdapter -InterfaceDescription "Microsoft KM-
    TEST Loopback Adapt"
Rename-NetAdapter -InterfaceDescription "Microsoft KM-TEST
    Loopback Adapter" -NewName "Loop1"
New-NetIPAddress -InterfaceIndex $adapter.ifIndex -AddressFamily
    "IPv4" -IPAddress "172.16.0.2" -PrefixLength 24
Route add 172.16.0.0 mask 255.240.0.0 172.16.0.1 metric 1
```

PowerShell – Allow SNMP Ports in Firewall

```
New-NetFirewallRule -DisplayName "SNMP" -Direction Inbound -
    Action Allow -Protocol UDP -LocalPort 161-162
```

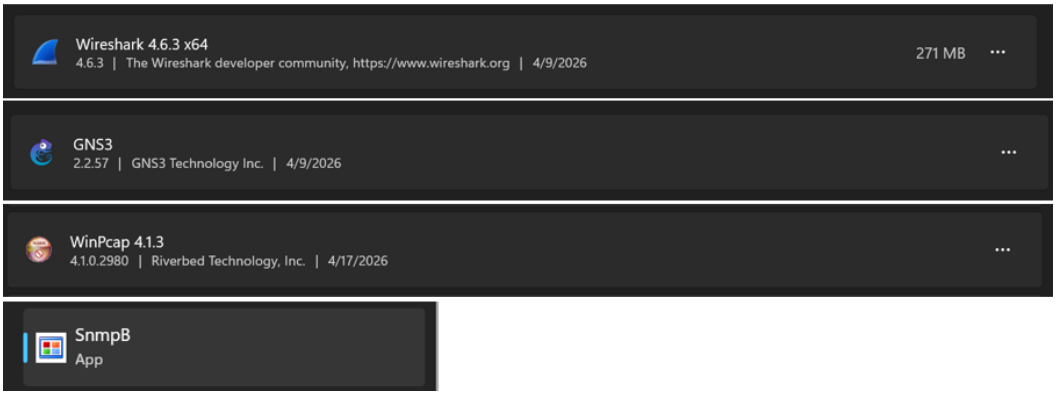
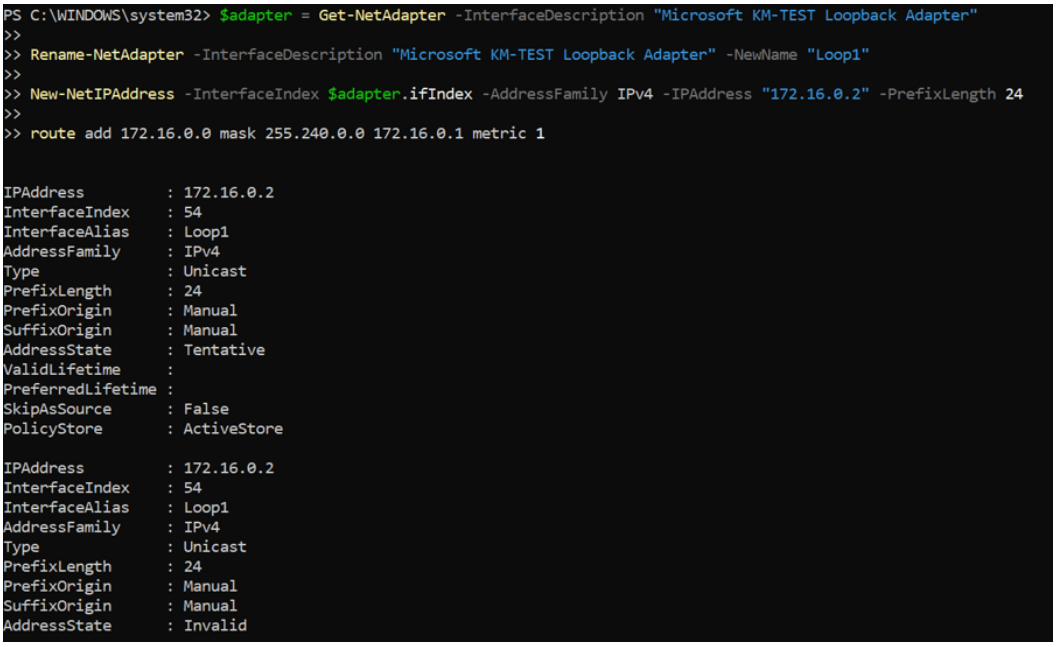


Figure 1: *Installed toolchain used across the lab environment*

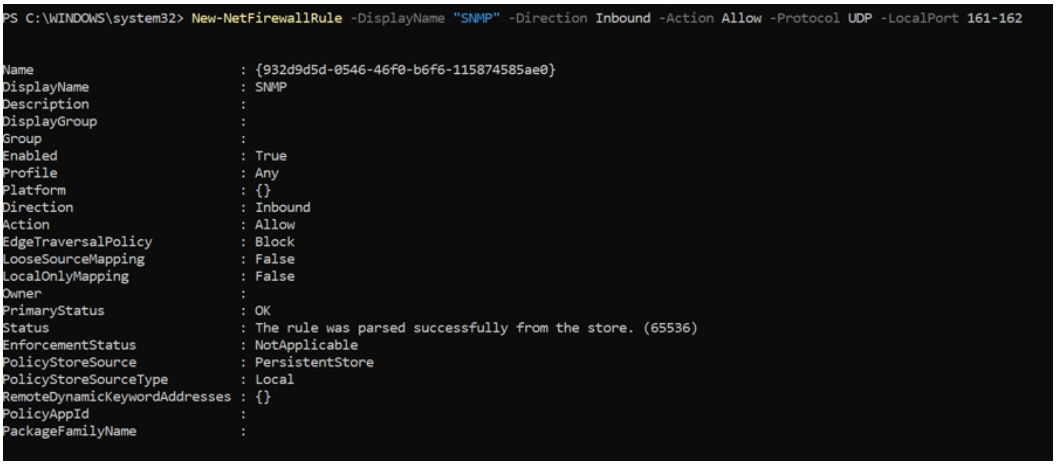


```
PS C:\WINDOWS\system32> $adapter = Get-NetAdapter -InterfaceDescription "Microsoft KM-TEST Loopback Adapter"
>>
>> Rename-NetAdapter -InterfaceDescription "Microsoft KM-TEST Loopback Adapter" -NewName "Loop1"
>>
>> New-NetIPAddress -InterfaceIndex $adapter.ifIndex -AddressFamily IPv4 -IPAddress "172.16.0.2" -PrefixLength 24
>>
>> route add 172.16.0.0 mask 255.240.0.0 172.16.0.1 metric 1


IPAddress         : 172.16.0.2
InterfaceIndex    : 54
InterfaceAlias    : Loop1
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 24
PrefixOrigin      : Manual
SuffixOrigin      : Manual
AddressState      : Tentative
ValidLifetime     :
PreferredLifetime :
SkipAsSource      : False
PolicyStore       : ActiveStore

IPAddress         : 172.16.0.2
InterfaceIndex    : 54
InterfaceAlias    : Loop1
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 24
PrefixOrigin      : Manual
SuffixOrigin      : Manual
AddressState      : Invalid
```

Figure 2: *Loopback adapter configuration on the host machine*



```
PS C:\WINDOWS\system32> New-NetFirewallRule -DisplayName "SNMP" -Direction Inbound -Action Allow -Protocol UDP -LocalPort 161-162


Name                          : {932d9d5d-0546-46f0-b6f6-115874585ae0}
DisplayName                   : SNMP
Description                   :
DisplayGroup                  :
Group                         :
Enabled                       : True
Profile                       : Any
Platform                      : {}
Direction                     : Inbound
Action                        : Allow
EdgeTraversalPolicy           : Block
LooseSourceMapping            : False
LocalOnlyMapping              : False
Owner                         :
PrimaryStatus                 : OK
Status                        : The rule was parsed successfully from the store. (65536)
EnforcementStatus             : NotApplicable
PolicyStoreSource             : PersistentStore
PolicyStoreSourceType         : Local
RemoteDynamicKeywordAddresses : {}
PolicyAppId                   :
PackageFamilyName             :
```

Figure 3: *Firewall rule permitting SNMP UDP traffic (161-162)*

3. Phase 1: SNMPv1/v2c Operations

3.1. SNMP Manager Setup (SnmpB)

A manager profile was configured to target R1 with community-based access.

Parameter	Value
Profile Name	R1
Agent Address	172.16.0.1
SNMP Version	SNMPv1 and SNMPv2c
Read Community	public

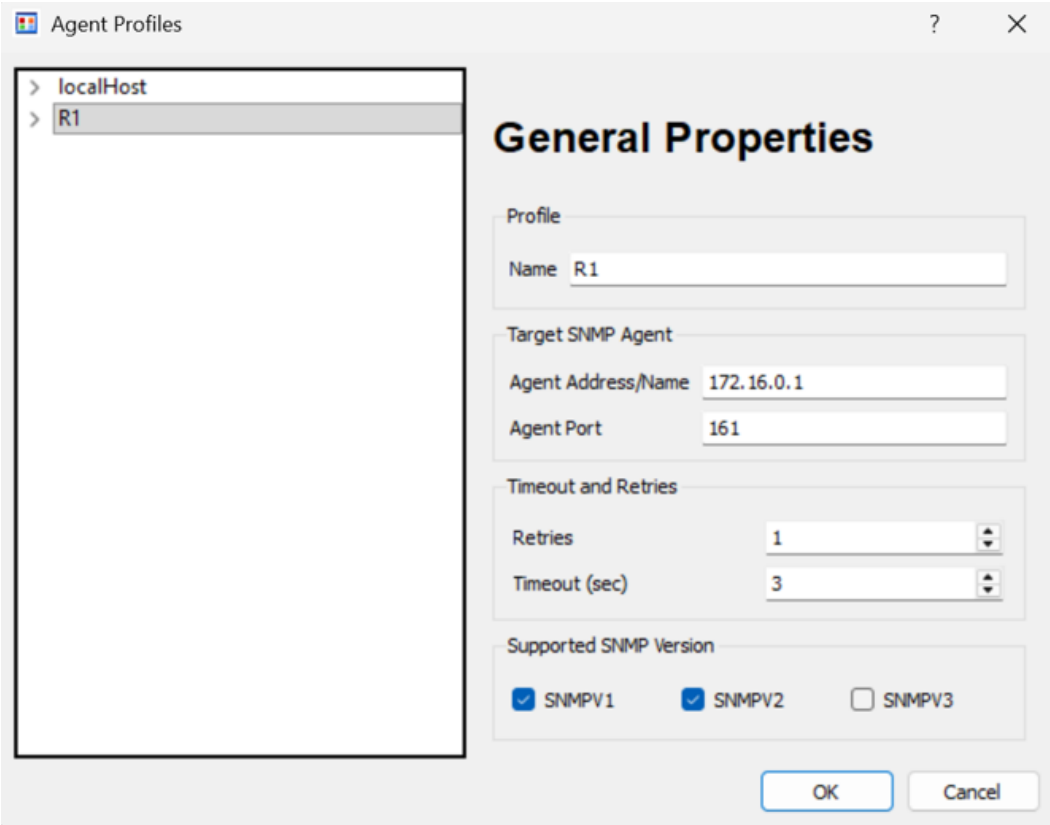


Figure 4: *SnmpB agent profile targeting router R1*

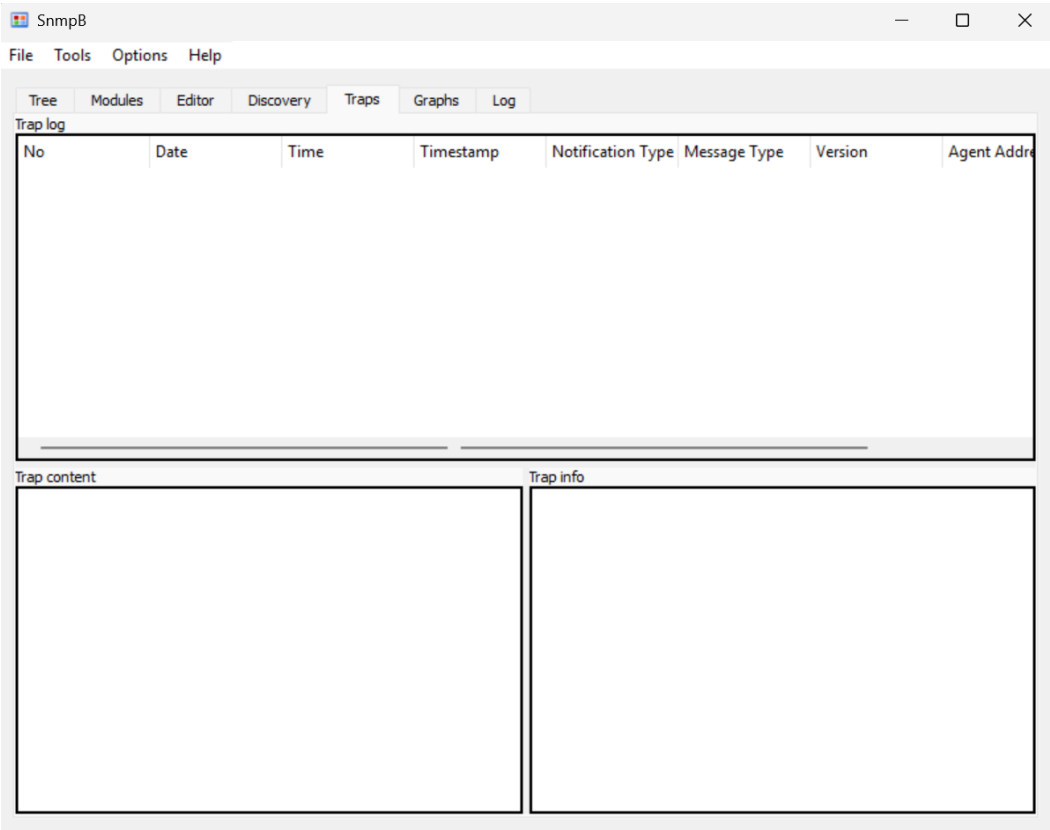


Figure 5: *SnmpB traps window listening for asynchronous notifications*

3.2. Cisco SNMP Agent Configuration

Cisco IOS – SNMPv1/v2c Baseline on R1

```
config t
snmp-server community public ro SNMP_ACL
snmp-server location lab-router-r1
snmp-server contact network-ops
snmp-server host 172.16.0.2 version 2c public
snmp-server enable traps
ip access-list standard SNMP_ACL
 permit 172.16.0.2
exit
```

Cisco IOS – Interface f0/0 Addressing

```
int f0/0
ip address 172.16.0.1 255.255.255.0
no shutdown
```

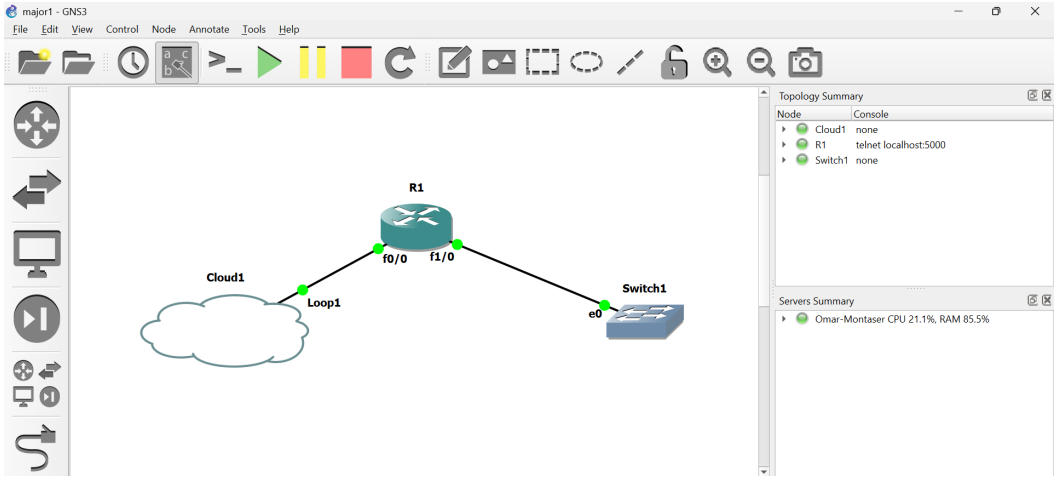


Figure 6: *GNS3 topology with direct manager-agent connectivity and active capture*

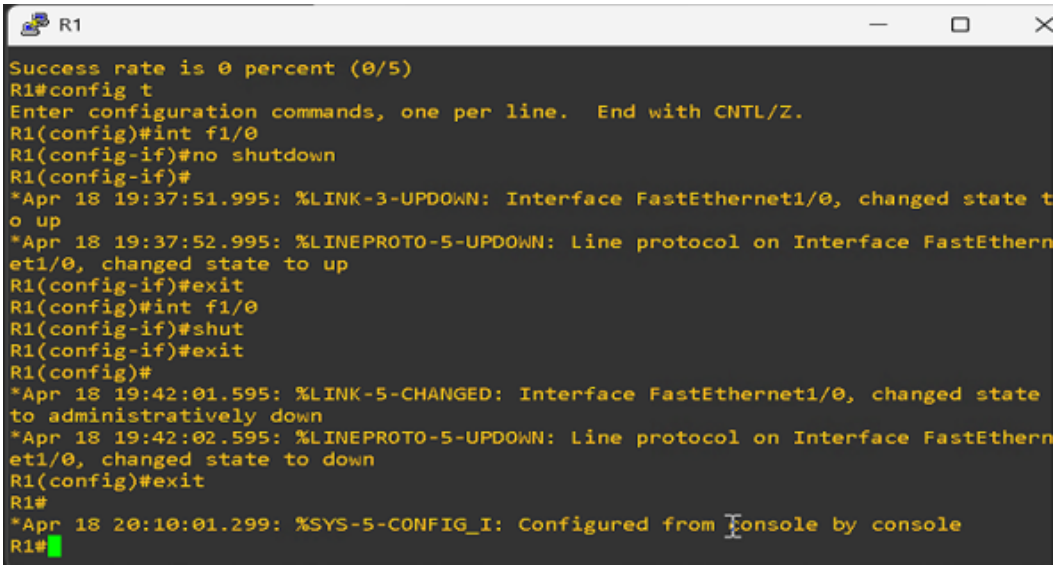
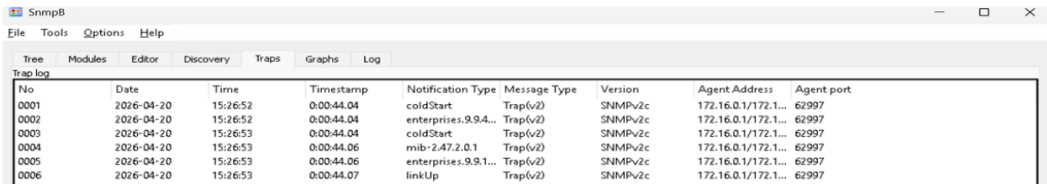


Figure 7: *Router console session used for SNMP configuration and validation*

3.3. Validation: Traps and MIB Walk

- Initial device state changes generated `coldStart` and `linkUp` traps.
- A walk over `mib-2.system` returned expected OIDs including `sysDescr`, `sysObjectID`, `sysUpTime`, `sysContact`, `sysName`, and `sysLocation`.



No	Date	Time	Timestamp	Notification Type	Message Type	Version	Agent Address	Agent port
0001	2026-04-20	15:26:52	0:00:44.04	coldStart	Trap(v2)	SNMPv2c	172.16.0.1/172.1...	62997
0002	2026-04-20	15:26:52	0:00:44.04	enterprises.9.9.4...	Trap(v2)	SNMPv2c	172.16.0.1/172.1...	62997
0003	2026-04-20	15:26:53	0:00:44.04	coldStart	Trap(v2)	SNMPv2c	172.16.0.1/172.1...	62997
0004	2026-04-20	15:26:53	0:00:44.06	mib-2.47.2.0.1	Trap(v2)	SNMPv2c	172.16.0.1/172.1...	62997
0005	2026-04-20	15:26:53	0:00:44.06	enterprises.9.9.1...	Trap(v2)	SNMPv2c	172.16.0.1/172.1...	62997
0006	2026-04-20	15:26:53	0:00:44.07	linkUp	Trap(v2)	SNMPv2c	172.16.0.1/172.1...	62997

Figure 8: *Initial SNMP traps received by the manager*

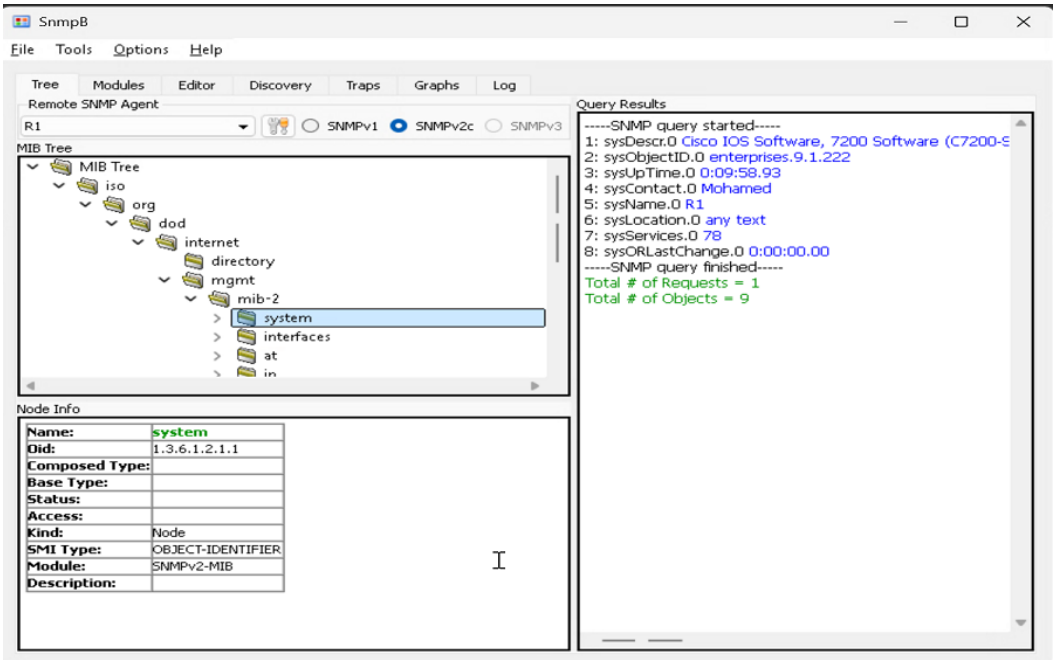


Figure 9: *Successful MIB walk over system subtree from SnmpB*

3.4. Protocol Behavior Comparison: SNMPv1 vs SNMPv2c

The same walk operation was repeated using SNMPv1 and SNMPv2c.

- **SNMPv1:** multiple `get-next-request/get-response` exchanges per OID.
- **SNMPv2c:** `GetBulkRequest` retrieves more data per transaction, reducing packet count and management overhead.

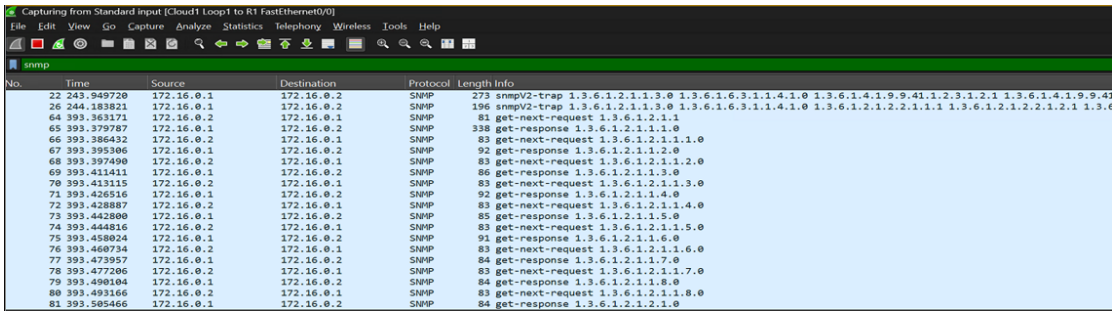


Figure 10: *SNMPv1 walk pattern with repeated get-next exchanges*

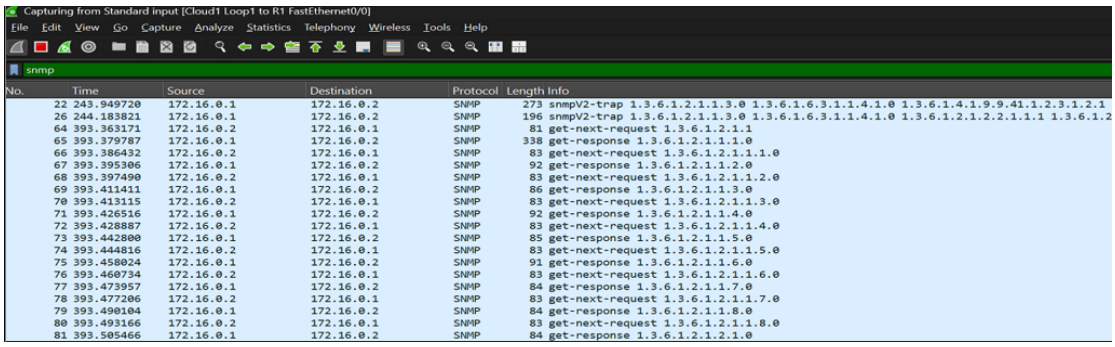


Figure 11: *SNMPv2c GetBulk behavior with fewer transactions*

3.5. Trap Generation via Interface State Changes

Cisco IOS – Force LinkUp Event

```
int f1/0
no shutdown
```

Cisco IOS – Force LinkDown Event

```
int f1/0
shutdown
```

Trap payloads included expected bindings such as `ifIndex`, `ifAdminStatus`, and `ifOperStatus`.

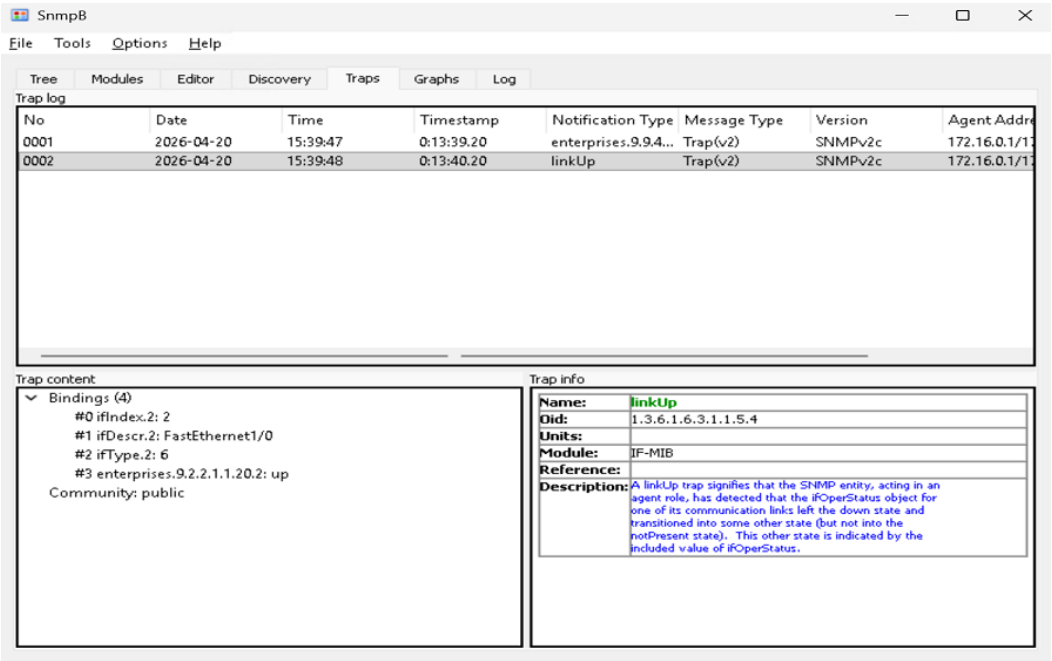


Figure 12: *LinkUp trap captured after enabling interface f1/0*

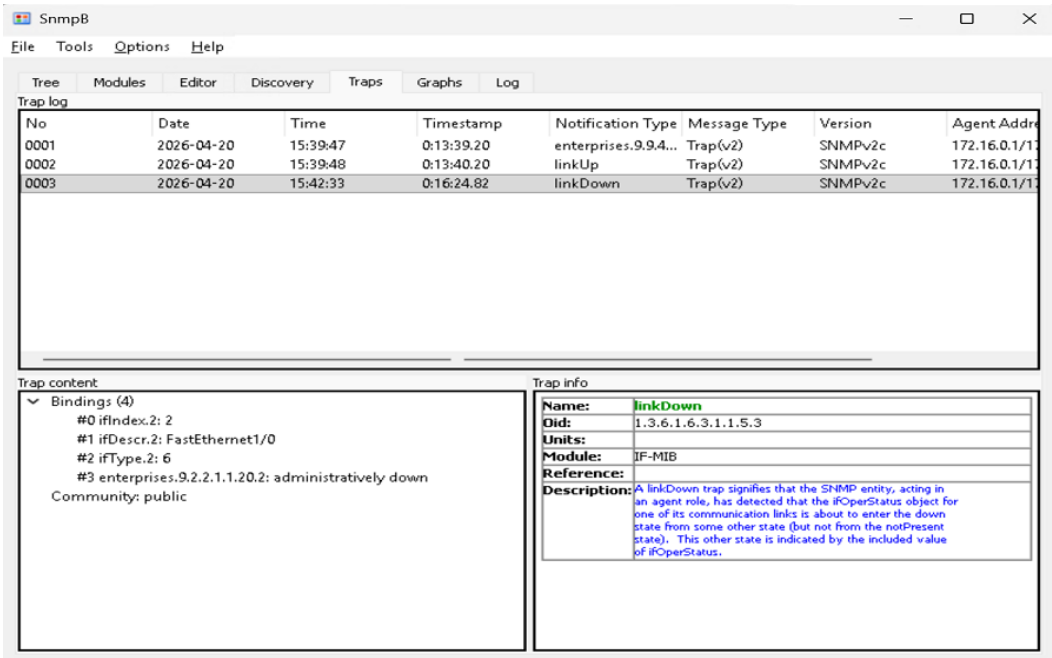


Figure 13: *LinkDown trap captured after shutting down interface f1/0*

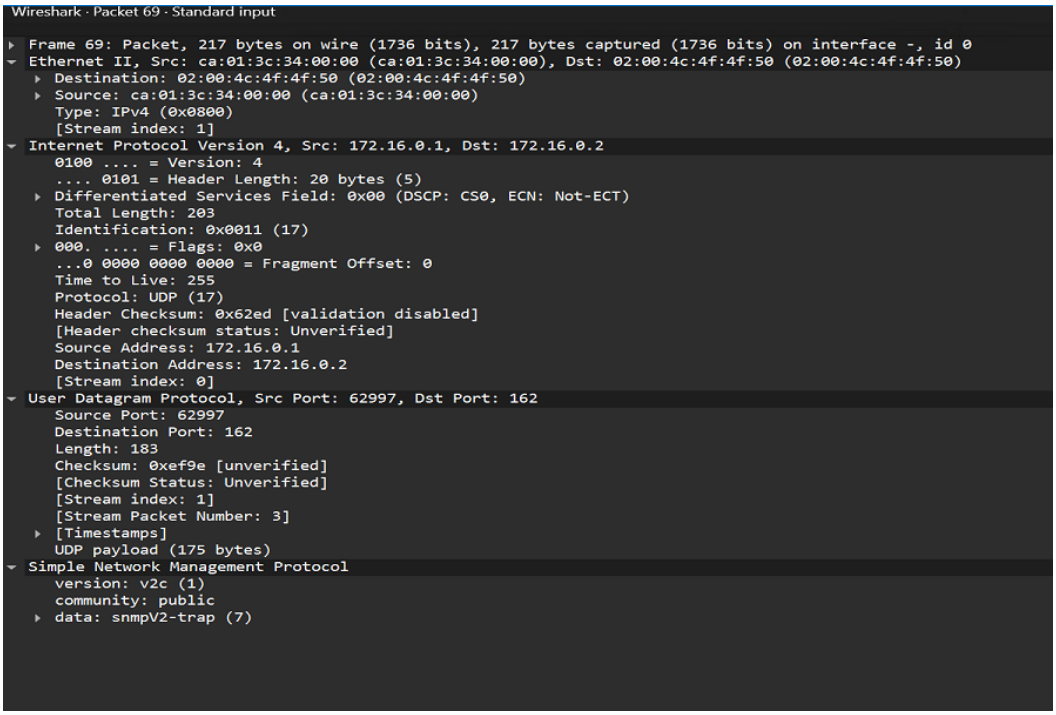


Figure 14: *Wireshark dissection of SNMP trap packet layers*

4. Phase 2: SNMPv3 Security Hardening

4.1. Design Goals

SNMPv3 was introduced to address the plaintext and weak-access limitations of SNMPv1/v2c by enforcing:

- **Authentication:** SHA-based integrity and source verification.
- **Privacy:** DES encryption for PDU confidentiality.
- **Authorization:** view/group/user-based access control.

4.2. SNMPv3 Configuration on Cisco Router

Cisco IOS – Engine ID

```
config t
snmp-server engineID local 123456789A
```

Cisco IOS – Views and Groups

```
snmp-server view VIEW_SYSTEM system included
snmp-server view VIEW_ALL iso included
snmp-server group GROUP1 v3 priv write VIEW_SYSTEM
snmp-server group GROUP2 v3 priv read VIEW_ALL
```

Cisco IOS – Users

```
snmp-server user USER1 GROUP1 v3 auth sha Auth1 priv des56 Enc1
snmp-server user USER2 GROUP2 v3 auth sha Auth2 priv des56 Enc2
end
```

Cisco IOS – Verification Commands

```
show snmp group
show snmp user
```

```
R1#show snmp group
groupname: ILMI                                security model:v1
readview : *ilmi                               writeview: *ilmi

notifyview: <no notifyview specified>
row status: active

groupname: ILMI                                security model:v2c
readview : *ilmi                               writeview: *ilmi

notifyview: <no notifyview specified>
row status: active

groupname: GROUP1                              security model:v3 priv
readview : v1default                           writeview: VIEW_SYSTEM

notifyview: <no notifyview specified>
row status: active

groupname: GROUP2                              security model:v3 priv
readview : VIEW_ALL                            writeview: <no writeview specified>

notifyview: <no notifyview specified>
row status: active

groupname: public                              security model:v1
readview : v1default                           writeview: <no writeview specified>

notifyview: <no notifyview specified>
row status: active      access-list: SNMP_ACL

groupname: public                              security model:v2c
readview : v1default                           writeview: <no writeview specified>

notifyview: *tv.FFFFFFFF.FFFFFFFF.FFFFFFFF.F
row status: active      access-list: SNMP_ACL

R1#
```

Figure 15: *SNMPv3 groups and permissions verified on router*

```
R1#show snmp user

User name: USER1
Engine ID: 123456789A
storage-type: nonvolatile        active
Authentication Protocol: SHA
Privacy Protocol: DES
Group-name: GROUP1

User name: USER2
Engine ID: 123456789A
storage-type: nonvolatile        active
Authentication Protocol: SHA
Privacy Protocol: DES
Group-name: GROUP2
```

Figure 16: *SNMPv3 users verified on router*

4.3. SNMPv3 Manager Configuration (SnmpB USM)

Two USM profiles and matching agent profiles were configured:

Parameter	USER1	USER2
Security Name	USER1	USER2
Authentication	SHA	SHA
Authentication Password	Auth1	Auth2
Privacy	DES	DES
Privacy Password	Enc1	Enc2

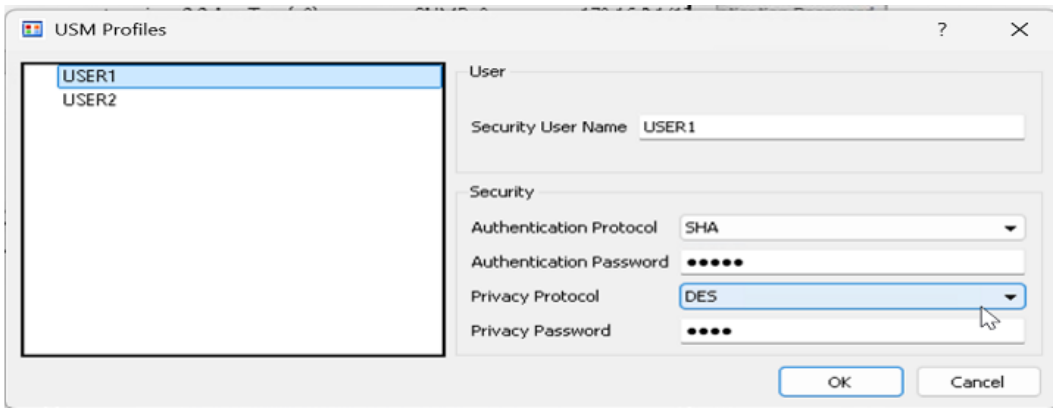


Figure 17: *USM profile for USER1 (authPriv)*

Figure 18: *USM profile for USER2 (authPriv)*

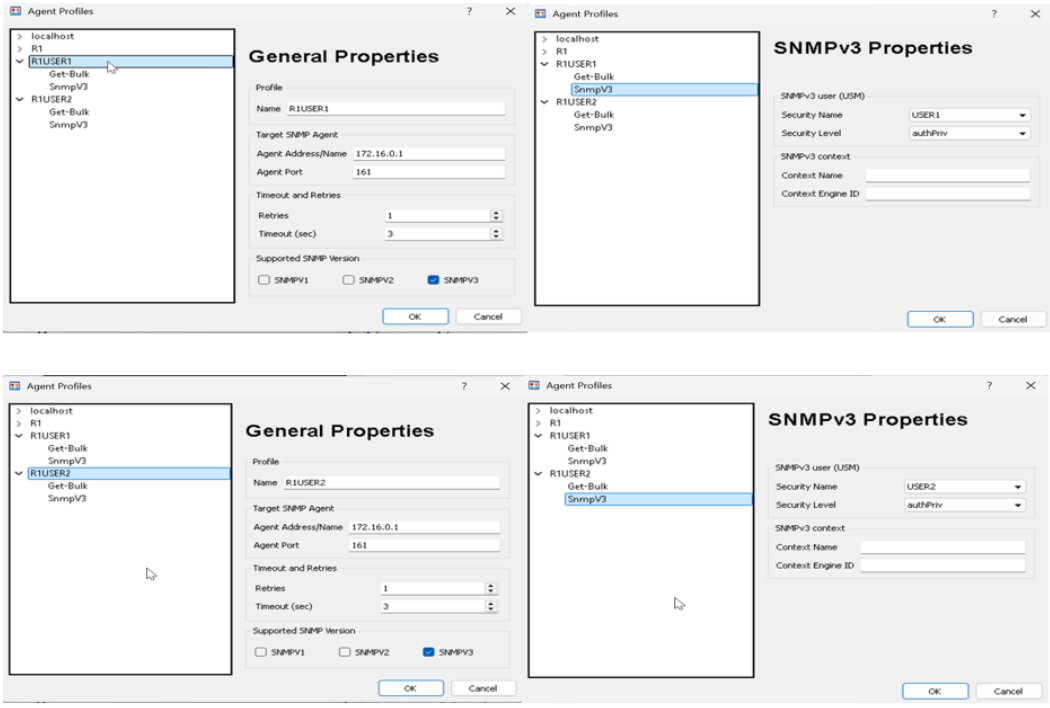


Figure 19: *Agent profiles mapped to SNMPv3 users*

4.4. Access Control Validation

- **USER1 (GROUP1, write VIEW_SYSTEM):** write operations on `sysContact` and `sysLocation` succeeded.
- **USER2 (GROUP2, read VIEW_ALL):** write attempt failed with authorization error, as expected.

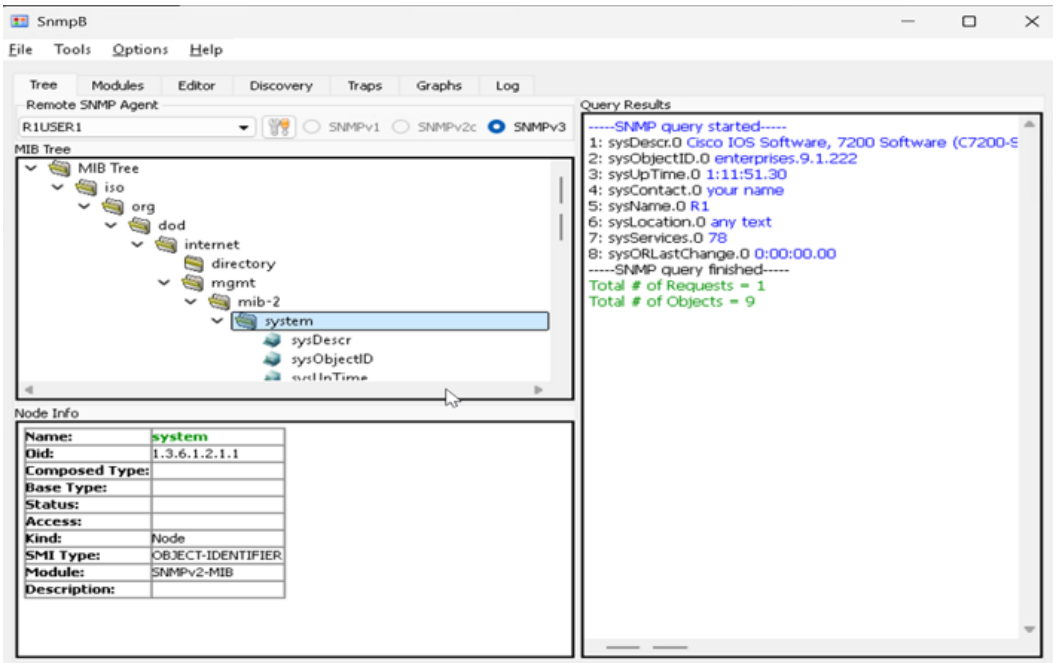


Figure 20: *MIB walk with USER1 credentials*

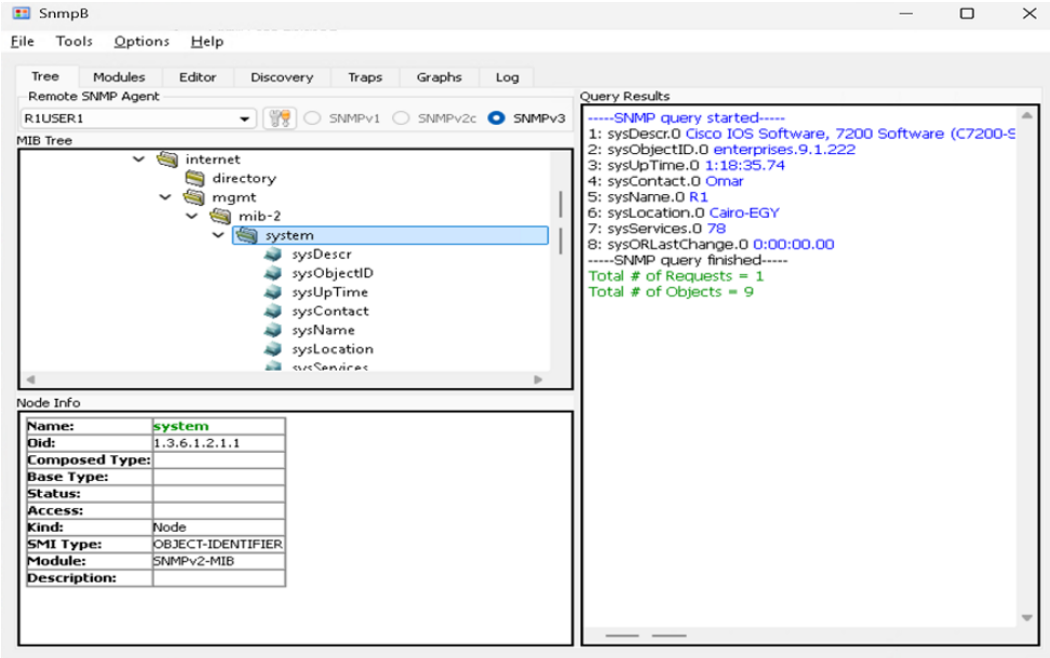


Figure 21: *Updated values after successful SNMP Set operations by USER1*

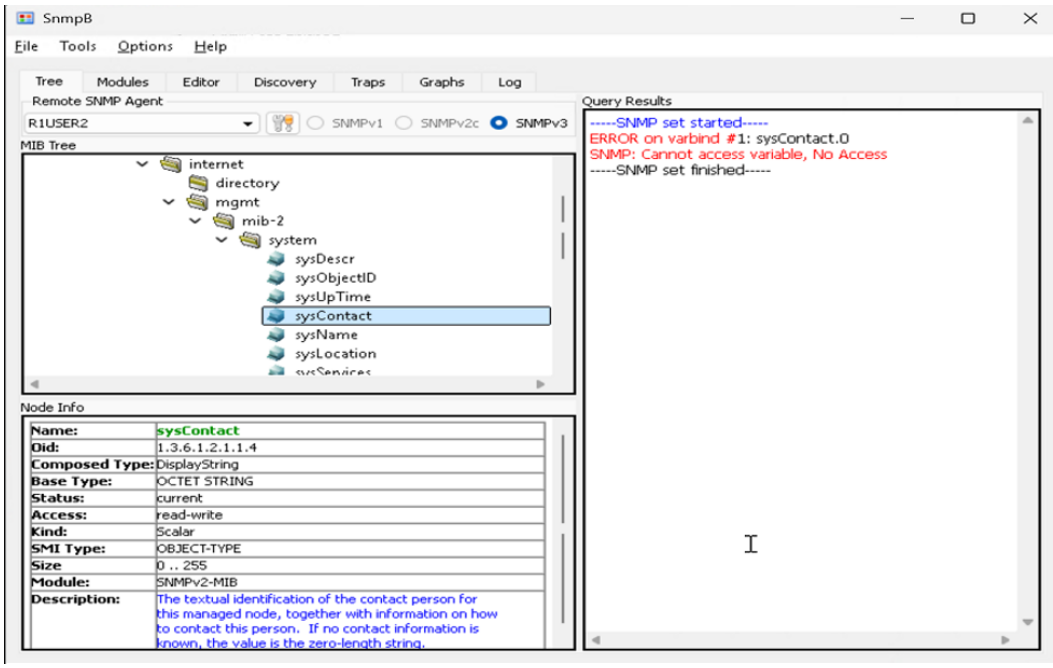


Figure 22: *Authorization failure when USER2 attempts a write operation*

5. Packet-Level Analysis in Wireshark

5.1. Encrypted SNMPv3 Traffic

A GetBulk request performed with SNMPv3 authPriv initially appeared as encrypted payload, and Wireshark reported unknown privacy key context.

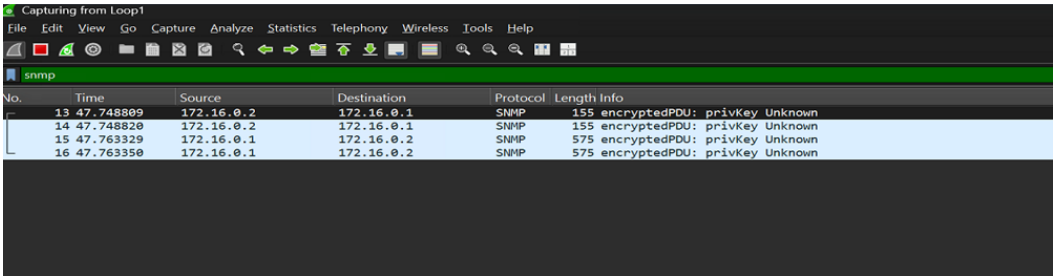


Figure 23: *SNMPv3 packets before providing user decryption keys*

5.2. Credential-Assisted Decryption

After adding SNMP users and their auth/privacy credentials to Wireshark's SNMP user table, packet contents were successfully decrypted.

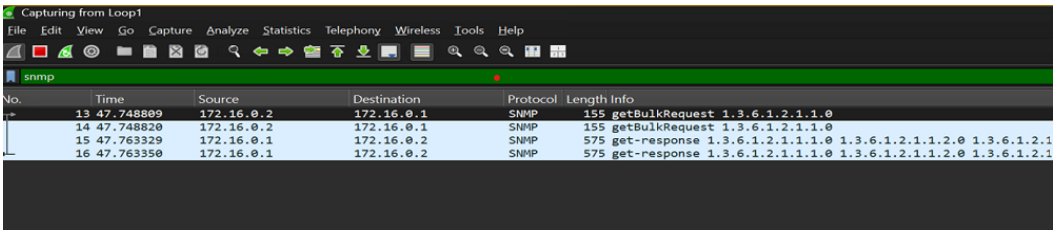


Figure 24: *SNMPv3 packet decryption after adding valid user credentials*

6. Key Outcomes

ID	Area	Outcome	Status
1	**Baseline connectivity**	Stable manager-agent communication established over loopback segment	✓
2	**Trap monitoring**	coldStart, linkUp, and linkDown traps captured and inspected	✓
3	**MIB operations**	Poll/walk operations completed for system subtree	✓
4	**Protocol comparison**	SNMPv2c GetBulk shown to reduce packet overhead vs SNMPv1	✓
5	**SNMPv3 security**	SHA + DES authPriv model deployed end-to-end	✓
6	**Access control**	Read/write separation validated through views, groups, and users	✓
7	**Traffic decryption**	Wireshark successfully decrypted SNMPv3 after key provisioning	✓

Technical Conclusions

- SNMPv1/v2c provides fast operational visibility but exposes management data in plaintext.
- SNMPv3 authPriv materially improves management-plane security by ensuring integrity and confidentiality.
- View/group/user mapping is effective for enforcing principle of least privilege in network monitoring environments.
- Packet analysis is essential for validating protocol behavior rather than relying on configuration state alone.

7. Future Improvements

- Migrate privacy from DES to AES where platform support is available.
- Add trap destination redundancy and SNMP manager high availability.
- Extend monitoring to additional MIB branches (interfaces, IP, CPU/memory).
- Integrate polling/trap data with a time-series dashboard stack for long-term observability.